

Applying B7's receiver gain to the beam scan — and why it makes the fit worse

Milestone. Aaron asked for rf-calibrator's per-cycle g_{rx} gain correction to be available in `beam_explorer.ipynb` as an interactive toggle, not baked in. The toggle is implemented (`apply_g_rx_cal (B7)`). This checkpoint records what it does, because the answer is not what the physics predicted and the default therefore needed justifying.

Question. Does dividing `measured_tx` by $g_{\text{rx}}(\nu, t)$ improve the beam fit?

Answer. No — it degrades it, substantially and consistently, and not because of interpolation error. Three controls below establish that the correction is implemented correctly and that the premise itself fails: the comb amplitude in this window does not scale with receiver gain.

Scope note. This dispatch was calibration only. The azimuth-zero / elevation-convention half is explicitly **not** touched — geometer's hardware-derived az-zero still tests $\sim 3\times$ worse than the free-fit optimum by both geometer's terrain MAD and my own beam-RMS check, and that discrepancy is back with geometer.

Inputs. `abscal/trx_phaseC.npz` (B7: 66 switching cycles, per-cycle $g_{\text{rx}}(\nu, t)$ and $T_{\text{rx}}(\nu, t)$, Phase C), `beam_explorer_cache.npz` (rebuilt 2026-09-17 with a `times` axis so the per-cycle solutions can be interpolated onto samples), `cal_gain.npz` (built by `build_cal_gain.py`).

Units. With the toggle off the measured quantity is raw correlator accumulator counts. With it on the quantity is counts / g_{rx} , so fit amplitudes change scale by $\sim 10^{-3}$; that is a unit change, not a result.

```
cache: beam_explorer_cache.npz
B7 calibration available: True
unsolved gap: 2026-07-17T19:42:14Z -> 2026-07-18T01:24:40Z
channels 101   samples 54240
```

1. Why this should have worked

`measured_tx` is the channel difference `auto[c] - ½(auto[c-1] + auto[c+1])`. With `auto = g_rx · (T_sky + T_rx)` and $g_{\rm rx}$ smooth across three adjacent 244 kHz channels, the difference carries $g_{\rm rx}$ as a **linear multiplicative factor** — the differencing removes the smooth bandpass, not the gain. So a drift in $g_{\rm rx}$ multiplies the data by a time-varying factor, and the model has exactly one free amplitude per channel, which cannot absorb it.

That drift is not small.

```
ch 712 (173.83 MHz), over used samples:
  g_rx 293.9 .. 473.2   ratio 1.61
```

```
across all 101 channels, the within-window g_rx drift ratio:
  median 1.66   range 1.47 .. 2.47
```

This is the standing ``receiver_regime_caveat`` in `beam_fits_v2`: a real, uncorrected multiplicative systematic on the amplitude scale.

2. What B7 covers here, and the one hole in it

B7 cycles: 66 total, spanning 2026-07-17 04:11:05.688671 -> 2026-07-18 02:57:01.180871 UTC
beam-scan window: 2026-07-17 18:48:52.993574 -> 2026-07-18 02:58:05.336946 UTC
-> the window sits ENTIRELY inside the calibrated span

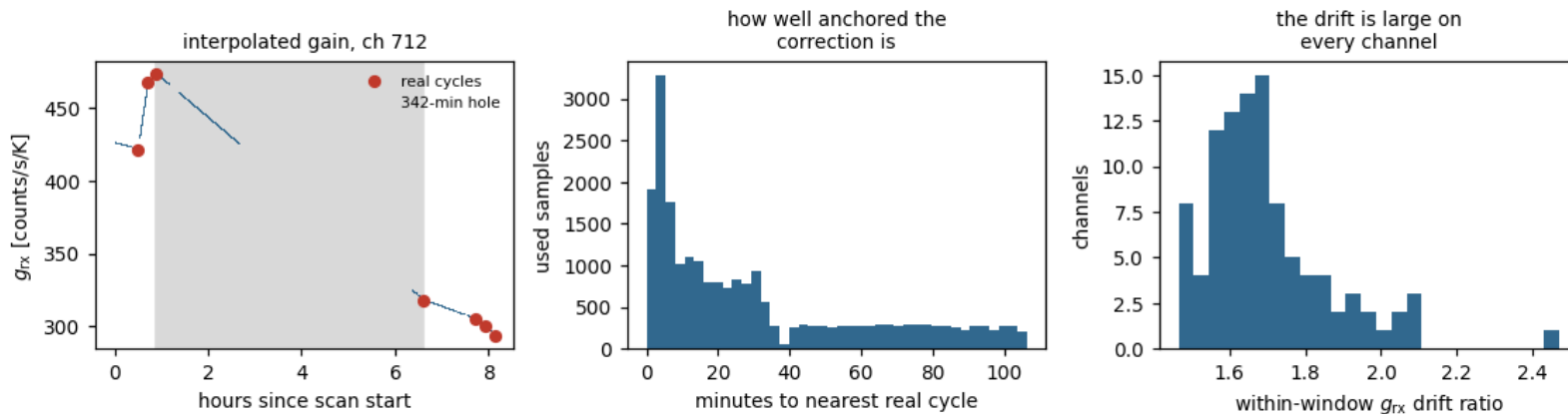
cycles inside the beam-scan window: 7
gaps between them [minutes]: [12.9 10.7 342.4 66.6 12.9 12.9]

used samples with a gain solution: 22614 of 22614
-> nothing is dropped. The 'trailing samples past the last cycle'
question turned out to be empty: no USED sample falls outside.

the 342-minute hole 2026-07-17T19:42:14Z -> 2026-07-18T01:24:40Z:
holds 10019 used samples (44.3% of them)
and it is exactly the interval B7 brackets the receiver regime change
to, so inside it the interpolation is a straight line through an
unknown transition -- not a measurement.

minutes from a used sample to the nearest real cycle:

p50	20.0
p75	50.3
p90	82.5
p95	94.2



3. The result: it degrades the fit

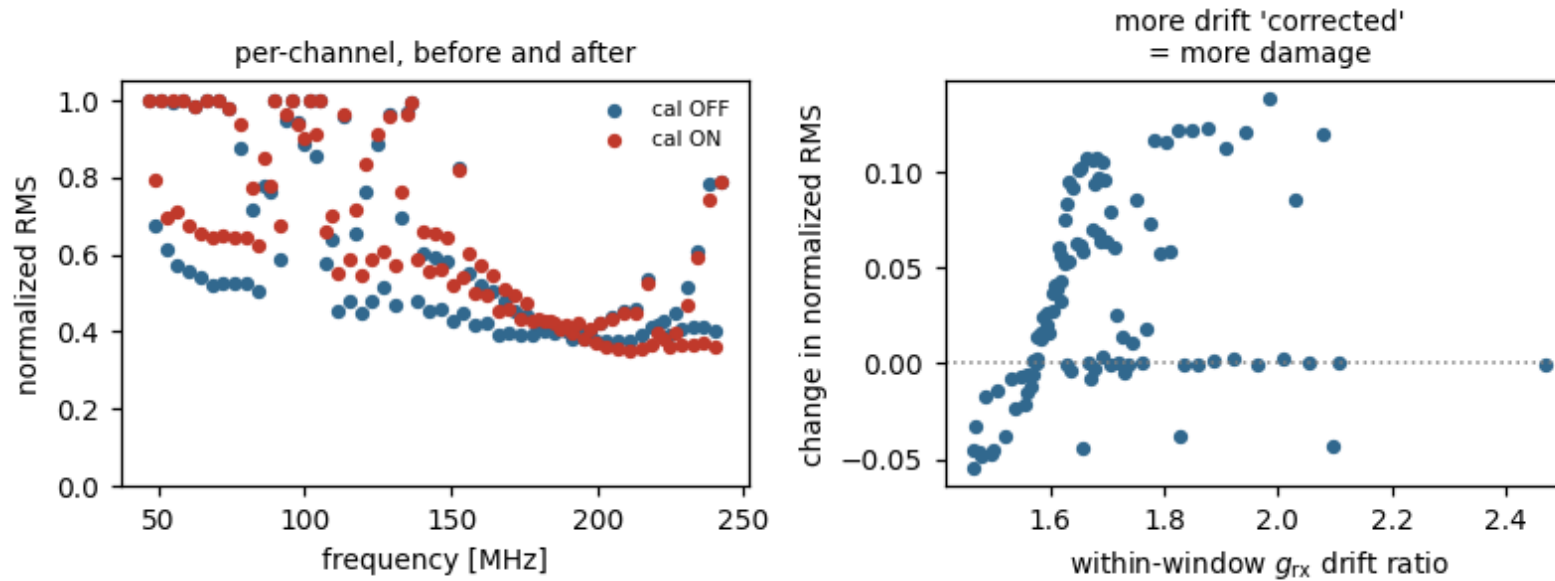
Pure HFSS (shape terms zero), amplitude refit in every case, so the only thing changing is the data.

```

ch 712, pure HFSS, amplitude refit
      normRMS      amplitude      n
cal OFF      0.3914      1.31e+11  22614
cal ON       0.4855      3.483e+08  22614
delta       +0.0941
all 101 channels, cal OFF: median 0.5176   below 0.5: 47   below 0.7: 72
all 101 channels, cal ON : median 0.5873   below 0.5: 37   below 0.7: 67

```

channels improved by the correction: 32/101 ; worsened: 69



`corr(drift ratio, RMS change) = +0.284`

4. Three controls

The obvious objection is that the correction is simply implemented wrong, or that the 342-minute interpolation is doing the damage. Both are testable.

CONTROL 1 -- constant gain must be a no-op

constant gain applied : 0.391375

no correction at all : 0.391375

difference : 1.1e-16

-> exact. The division, the masking and the amplitude refit are correct, and only the TIME VARIATION of g_rx can matter.

CONTROL 2 -- by distance to the nearest real cycle

subset	n	cal OFF	cal ON	delta
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anchor < 15 min	9778	0.4093	0.5578	+0.1485
-----------------	------	--------	--------	---------

anchor < 30 min	14373	0.3960	0.5173	+0.1212
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anchor > 60 min	4646	0.3816	0.3125	-0.0692
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-> the correction hurts MOST where g_rx is best known, and HELPS where it is least known. That is backwards for an interpolation-error explanation.

CONTROL 3 -- divide the data by g_rx**p

p	normRMS
---	---------

-1.00	0.3472
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-0.50	0.3603
-------	--------

-0.25	0.3738
-------	--------

+0.00	0.3914
-------	--------

+0.25	0.4121
-------	--------

+0.50	0.4353
-------	--------

+1.00	0.4855
-------	--------

+2.00	0.5842
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-> monotonic across the whole range. The fit prefers MULTIPLYING by g_rx (p = -1, RMS 0.347) to leaving it alone (p = 0, 0.391) to dividing by it (p = +1, 0.486). There is no interior optimum at p > 0, which is what a genuine gain correction would produce.

5. What this establishes, and what it does not

Established.

- The correction is implemented correctly. Control 1 is exact to 10^{-16} .
- Applying it degrades the fit: ch 712 pure-HFSS **0.3914 → 0.4855**; fleet median **0.5176 → 0.5873**; **69 of 101** channels worse; and the damage scales with how much g_{rx} drifts on that channel.
- The 342-minute interpolation hole is **not** the cause. Control 2 shows the damage is largest on the best-anchored samples and the correction helps on the worst-anchored ones — the opposite of interpolation error.
- Control 3 is monotonic with no interior optimum, so the data carry no positive exposure to g_{rx} at all.

So the premise — that the beam-scan comb amplitude scales as g_{rx}^{+1} — is not supported.

A tempting inference, which I am deliberately not making. A far-field transmitter signal received through the antenna and the full analog chain must scale with g_{rx} . A tone injected downstream of the LNA — the digital self-comb at 250/128 MHz that `load_v007_data`'s own docstring and `flags/v0` both point to — would not. That makes this look like evidence on the open TX-identity dispute.

This test cannot carry that weight, for one specific reason. The beam scan is **time-ordered**: pointing and time are entangled, so any time-varying quantity correlates with the pointing pattern — and the pointing pattern is the signal being fit. The raw $\text{corr}(g_{\text{rx}}, |\text{measured}_{\text{tx}}|) = -0.33$ is therefore confounded and must not be read as receiver physics. A common thermal driver moving both LNA gain and radiated comb amplitude would produce the same signature.

Deferred findings

- Settling the above needs a **pointing-controlled** test: regress comb amplitude against g_{rx} within fixed (az, el) cells, so the beam pattern is held constant and only the time variation is free. Not attempted here; it is a separate piece of work with a real bearing on TX identity.
- T_{rx} from B7 is loaded into `cal_gain.npz` (`trx_cycles`) but not used. A T_{rx} -based correction is a different model — additive, not multiplicative — and was not part of this dispatch.

- The $\times 1.61 g_{\rm rx}$ drift remains a real, uncorrected systematic on the absolute amplitude scale. Nothing here fixes it; it establishes that dividing it out does not help the shape fit.

Decision requested

1. **Keep the toggle defaulting to OFF.** Applying the correction as specified makes the fit worse on 69 of 101 channels, and the controls show that is not a bug. I recommend off, with the toggle there for inspection.
2. **Authorize (or decline) the pointing-controlled $g_{\rm rx}$ regression.** It is the test that would turn this anomaly into evidence on TX identity, which is the largest open question on this dataset.
3. Whether to hand this to `rf-calibrator` — the finding is about how their product behaves against the beam data, and they may have an explanation (e.g. a known injection point for the comb) that closes it immediately.

STOPPED AT REVIEW GATE — awaiting Aaron's approval.